PRACTICE PROBLEMS (NM)

Date:

BISECTION METHOD: (chap # 2)

Q₁: Appr. the smallest +ve root of $f(x) = e^{-x} - x$ in the interval $[0, 1]$ using the bisection method correct upto 3 decimal places.

$\overset{a}{[0},\overset{b}{1]}$

Sol. $f(x) = e^{-x} - x$ $\quad f(a) = f(0) = 1$ $\quad f(b) = f(1) = -0.63212.$

$f(a) = +ve$; $f(b) = -ve$; $f(x) = -ve \to b : x$ $\quad f(b) = f(x)$

$x = \frac{0+1}{2} = 0.5$ $\quad f(x) = +ve \to a : x$ $\quad f(a) = f(x)$

Itr no.	a	b	x	f(a)	f(b)	f(x)
1	0	1	0.5	1	-0.63212	0.10653
2	0.5	1	0.75	0.10653	-0.6321	-0.27763
3		0.75	0.625		-0.27763	-0.08974
4		0.625	0.5625		-0.08974	0.00728
5	0.5625		0.59375	0.00728		-0.04149
6		0.59375	0.57812		-0.04149	-0.01717
7		0.57812	0.57031		-0.01717	-4.95×10^{-3}
8		0.57031	0.56641		-4.95×10^{-3}	1.1493×10^{-3}
9.	0.56641		0.56836	1.14×10^{-3}		-1.9063×10^{-3}
10		0.56836	0.56738		-1.9×10^{-3}	-3.7×10^{-4}
11		0.56738	0.56690		-3.7×10^{-4}	3.81×10^{-4}
12	0.56690		0.56714	3.8×10^{-4}		5.15×10^{-4}
13	0.56714		0.567			

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Q2: Solve for the root of $f(x) = x^3 - 7x + 6$ in the interval $[1.5, 2.4]$ using bisection method, correct upto 3 decimal places.

Sol: $f(a) = f(1.5) = -1.125$ $\quad$ $f(b) = f(2.4) = 3.024$

Note: Sign remains same for each $f(k)$. Change as per this

itr	a	b	x	f(a)	f(b)	f(x)
1	1.5	2.4	1.95	(−) 1.125	(+) 3.024	−0.23512
2	1.95		2.175	(−) 0.23512		1.06411
3		2.175	2.0625		(+) 1.06411	0.33618
4		2.0625	2.00625		0.33618	0.03148
5		2.00625	1.97812		0.03148	−0.10654
6	1.97812		1.99219	−0.10654		−0.03868
7	1.99219		1.99922	−0.03868		-3.89×10^{-3}
8	1.99922		2.00273	-3.89×10^{-3}		0.01369
9		2.00273	2.00098		0.01369	4.9×10^{-3}
10		2.00098	2.00009			

Q3. $f(x) = \tan x - x$ $\quad$ $[1.5, 2.4]$ $\quad$ correct upto 3 decimal places.

itr	a	b	x	f(a)	f(b)	f(x)
1	1.5	2.4	1.95	12.601	−3.316	−4.45948
2		1.95	1.725		−4.459	−8.15
3		1.725	1.6125		−8.15	−25.5
4		1.6125	1.55625		−25.5	67.18
5	1.55625		1.58437	67.18		−75.2
6		1.58437	1.57031		−75.2	2054.66
7	1.57031		1.57734	2054.66		−154.39

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	a	b	x	f(a)	f(b)	f(x)
8	1.57031	1.57734	1.57382	2054.66	-154.39	-332.2
9		1.57382	1.57207		-333.2	-786.7
10		1.57207	1.57119		-786.7	-2541.7
11		1.57119	1.57075		-2541.7	21584.2
12	1.57075		1.57097	21584.2		

NEWTON-RAPHSON :

Q1 Find the root of $f(x)=x^2-2$ using the N.R method starting at $x_0=1$

$f(x)=x^2-2$

$f'(x)=2x$

$f(x_0)=f(1)=-1$

$f'(x_0)=f'(1)=2$

$x_{i+1} = x_i - \frac{f(x_0)}{f'(x_0)}$

i	x_{i+1}
0	1.5
1	1.41667
2	1.41421
3	1.41421

→ exactly similar = Ans

Q2: $f(x)=x^3-2x+1$ $x_0=0$

$f'(x)=3x^2-2$

$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$

i	x_{i+1}
0	$x_1 = 0.5$
1	~~0.75~~ ~~1~~ 0.6
2	~~1.0625~~ 0.61739
3	0.61803
4	0.61803

Q3 : $f(x)=\cos(x)-x$ $x_0=0$

$f'(x)= -\sin(x)-1$

i	x_{i+1}
0	1
1	0.75036
2	0.7391
3	0.739085
4	0.739085

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Q4. $f(x) = e^x - 3x \quad x_0 = 1$

$f'(x) = e^x - 3$

$$x_{i+1} = x_i - \frac{e^{x_i} - 3x_i}{e^{x_i} - 3}$$

i	x_{i+1}
0	0
1	0.5
2	0.61006
3	0.618996
4	0.61906
5	0.61906

Q5. $f(x) = x^3 - 4x^2 + 6 \quad x_0 = 2$

$f'(x) = 3x^2 - 8x$

$$x_{i+1} = x_i - \frac{x_i^3 - 4x_i^2 + 6}{3x_i^2 - 8x_i}$$

i	x_{i+1}
0	1.5
1	1.57143
2	1.57199
3	1.57199

SECANT METHOD:

Q1: Find the root of $f(x) = x^3 - x - 2$ using the secant method with the initial guesses $x_0 = 1$ and $x_1 = 2$ correct upto 4 decimal places.

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad x_{-1} \quad\quad\quad\quad x_i$

$f(x_0) = \cancel{x_{i-1}} = 1$; $\cancel{x_i} = 2$, $x_{i+1} = x_i - \dfrac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$

~~f(x0) = f(~~

$f(x_0) = -2 \quad f(x_1) = 4 \quad x_0 = 1 \quad x_1 = 2.$

$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$= 2 - \frac{4(1-2)}{(-2) - (4)}$$

$\boxed{x_2 = 1.333}$

$$x_3 = 1.3333 - \frac{(-0.96310)(2 - 1.3333)}{(4) - (-0.96310)}$$

$\boxed{x_3 = 1.46267}$

$$x_4 = 1.46267 - \frac{(-0.33343)(1.3333 - 1.46267)}{(-0.96310) - (-0.33343)}$$

$\boxed{x_4 = 1.53118}$

$$x_5 = 1.53118 - \frac{(0.05869)(1.46267 - 1.53118)}{(-0.33343) - (0.05869)}$$

$x_6 =$

$$x_6 = 1.52681 - \frac{(0.03235)\cdot(1.53118 - 1.52681)}{(0.05869) - (0.03235)}$$

$\therefore x_{i-1} = 0$ (min. given)

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i	x_{i-1}	x_i	$f(x_{i-1})$	$f(x_i)$	x_{i+1}
1	1	2	-2	4	1.33333
2	2	1.3333	4	-0.96310	1.46267
3	1.3333	1.46267	-0.96310	-0.33343	1.53118
4	1.46267	1.53118	-0.33343	0.05869	1.526801
5	1.53118	1.526801	0.05869	0.03235	1.520681
6	1.52681	1.520681	0.03235	-0.00415	1.51978
7	1.520681	1.519078	-0.00415	-0.009497	1.52138
8	1.51978	1.52138	-0.009497	0.0000	1.52138

Ans

$$x_7 = 1.520681 - \frac{(-0.00415)(1.52681 - 1.520681)}{(0.03235) - (-0.00415)}$$

$$x_8 = 1.51978 - \frac{(-0.009497)(1.520681 - 1.51978)}{(-0.00415) - (-0.009497)}$$

$x_9 = 1.5\cancel{1978}^{2138}$ $\underline{(0.0000)}$

Q. Determine the root of $f(x) = \cos x - x$ correct upto 3 decimal places:
$x_0 = 0 \quad x_1 = 1$

$$x_2 = 1 - \frac{(-0.459697)(0-1)}{(1) - (-0.459697)}$$

$x_2 = 0.68507$

$$x_3 = 0.68507 - \frac{(0.08930)(1 - 0.68507)}{(-0.459697) - (0.08930)}$$

$x_3 = 0.736296$.

$$x_4 = 0.736296 - \frac{(0.00466)(0.68507 - 0.73630)}{(0.08930) - (0.00466)}$$

$x_4 = 0.73912$

$$x_5 = 0.73912 - \frac{(-5.83539)(0.73630 - 0.73912)}{(0.00466) - (-5.83539)}$$

$= 0.73630$

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i	x_{i-1}	x_i	$f(x_{i-1})$	$f(x_i)$	x_{i+1}
1	0	1	1	-0.459697	0.68507
2	1	0.68507	-0.459697	0.08930	0.73630
3	0.68507	0.73630	0.08930	0.00466	0.73912
4	0.73630	0.73912	0.00466	-5.83539	0.73630
5	0.73912	0.73630	-5.83539	0.00242	0.73630

$$x_6 = 0.73630 - \frac{(0.00242)(0.73912 - 0.73630)}{(-5.83539) - (0.00242)} = 0.73630$$

Q_3 locate root of $f(x) = e^{-x} - x$ using secant method $x_0 = 0$ $x_1 = 1$ correct upto 4 decimal places

1	0	1	1	-0.63212	0.61270
2	1	0.61270	-0.6321	-0.07081	0.56384
3	0.61270	0.56384	-0.07081	0.00518	0.56717
4	0.56384	0.56717	0.00518	-4.1×10^{-5}	0.56714

$$\cancel{x_6 = 0.73630 - \frac{(0.00242)(0.73912 - 0.73630)}{(-5.83539) - (0.00242)} = 0.73630}$$

Rem part of Q4.

$$x_5 = 0.24914 - \frac{(0.01290)(0.21739 - 0.24914)}{(0.14068) - (0.01290)}$$

25472 0.24914 25472 -0.00135 -0.0027

$$x_2 = 1 - \frac{(-2)(0-1)}{(1)-(-2)} = 0.33333$$

$$x_3 = 0.3333 - \frac{(-0.29628)(1-0.33)}{(-1) - (-0.29628)} = 0.21989$$

$$x_4 = 0.21739 - \frac{(0.14068)(0.33 - 0.21739)}{(-0.29628) - (0.14068)}$$

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$$x_2 = 1 - \frac{(-0.63212)(0-1)}{-(-0.63212)+1} = 0.61270$$

$$x_3 = 0.61270 - \frac{(-0.07081)(1-0.61270)}{(-0.63212)-(-0.07081)} = 0.56384$$

$$x_4 = 0.56384 - \frac{(0.00518)(0.61270-0.56384)}{(-0.07081)-(0.00518)} = 0.56717$$

$$x_5 = 0.56717 - \frac{(-4.1\times10^{-5})(0.56384\cancel{8}-0.56717)}{(0.00518)-(-4.1\times10^{-5})} = 0.56714$$

Q4. $f(x) = x^3 - 4x + 1$ $x_0 = 0$ $x_1 = 1$ correct upto 5 decimal places

1	0	1	1	−2	0.33333
2	1	0.33333	−2	−0.29628	0.21739
3	0.33333	0.21739	−0.29628	0.14068	0.~~24914~~ 25472
4	0.21739	0.~~24914~~ 25472	0.14068	~~0.01890~~ −0.00235	0.25410
5	0.25472	0.25410	−0.00235	0.0000	0.25410

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GUASS SIEDAL METHOD.

Q1. In a distributed computer system, three servers S_1, S_2 & S_3. share processing load.
Because of data dependency & task migration the effective load on each server depends on other.
The load balance equations are:

$$10S_1 - 2S_2 - S_3 = 27$$
$$-3S_1 + 9S_2 + 2S_3 = -22.$$
$$-2S_1 - S_2 + 7S_3 = 14$$

Apply Guass Siedal method to find server's load.

Solution. $x_3^{(0)} = x_2^{(0)} = x_1^{(0)} = 0$

Dominancy check:

$|10| > |2| + |1|$, $|9| > |2| + |-3|$, $|7| > |-1| + |-2|$

$x_1 = \frac{1}{10}[27 + 2x_2 + x_3]$, $x_2 = \frac{1}{9}[-22 + 3x_1 - 2x_3]$, $x_3 = \frac{1}{7}[14 + 2x_1 + x_2$

Itr.	x_1	x_2	x_3
1	2.7000	-1.5444	2.5508
2	2.6462	-2.129	2.5419
3	2.5193	-2.1495	2.4127
4	2.5114	-2.1435	2.4113
5	2.5124	-2.1428	2.4117
6	2.5126	-2.1428	2.4118
7	2.5126	-2.1428	2.4118

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Q2. A computer network consist of 3 routers R_1, R_2 and R_3, due to packet forwarding & congestion effects the traffic from each router depends on traffic from other router. The steady state traffic equation in Mbps.

$$12R_1 - 2R_2 + R_3 = 19$$
$$-R_1 + 10R_2 - 2R_3 = 17$$
$$2R_1 - R_2 + 8R_3 = 25$$

$$\frac{1}{12}[19 + 2R_2 - R_3]$$
$$\frac{1}{10}[17 + R_1 + 2R_3]$$
$$\frac{1}{8}[25 - 2R_1 + R_2]$$

Itr	R_1	R_2	R_3
0	0	0	0
1	1.583	1.858	2.961
2	1.646	2.466	3.020
3	1.741	2.478	2.999
4	1.746	2.475	2.999
5	1.745	2.474	2.997
6	1.745	2.474	2.997

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CHOLESKEY METHOD:

γ of book (7)

Q_1: $9x_1 + 6x_2 + 12x_3 = 17.4$

$6x_1 + 13x_2 + 11x_3 = 23.6$

$12x_1 + 11x_2 + 26x_3 = 30.8.$

Sol

$$A = \begin{bmatrix} 9 & 6 & 12 \\ 6 & 13 & 11 \\ 12 & 11 & 26 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 17.4 \\ 23.6 \\ 30.8 \end{bmatrix}$$

$A = L \cdot L^T$

$$A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} \Rightarrow A = \begin{bmatrix} l_{11}^2 & l_{21}l_{11} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

A/ = Comparing Values:

$l_{11}^2 = 9 \Rightarrow \boxed{l_{11} = 3}$, $l_{11}l_{31} = 12 \Rightarrow (3)l_{31} = 12 \Rightarrow \boxed{l_{31} = 4}$

$l_{21}l_{11} = 6 \Rightarrow l_{21}(3) = 6 \Rightarrow \boxed{l_{21} = 2}$

$\cancel{l_{11}l_{21} = 6} \Rightarrow l_{21}^2 + l_{22}^2 \Rightarrow 13 \Rightarrow 4 + l_{22}^2 = 13 \Rightarrow \boxed{l_{22} = 3}$

$l_{21}l_{31} + l_{22}l_{32} = 11 \Rightarrow (2)(4) + (3)l_{32} = 11 \Rightarrow \boxed{l_{32} = 1}$

$l_{31}^2 + l_{32}^2 + l_{33}^2 = 26 \Rightarrow (4)^2 + (1)^2 + l_{33}^2 = 26. \Rightarrow \boxed{l_{33} = 3}$

Forward Substitution:

$LY = b$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 17.4 \\ 23.6 \\ 30.8 \end{bmatrix}$$

$l_{31}y_1 + l_{32}y_2 + l_{33}y_3 = 30.8$

$4(5.8) + (1)(4) + 3y_3 = 30.8$

$\boxed{y_3 = 1.2}$

$l_{11}y_1 = 17.4 \Rightarrow (3)y_1 = 17.4 \; \boxed{y_1 = 5.8}$

$l_{21}y_1 + l_{22}y_2 = 23.6 \Rightarrow 2(5.8) + 3y_2 = 23.6 \Rightarrow \boxed{4 = y_2}$

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Backward Substitution:

$L^T X = y$

$$\begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.8 \\ 4 \\ 1.2 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.8 \\ 4 \\ 1.2 \end{bmatrix}$$

$3x_3 = 1.2 \Rightarrow \boxed{x_3 = 0.4}$

$3x_2 + x_3 = 4 \Rightarrow \boxed{x_2 = 1.3}$

$3x_1 + 2x_2 + 4x_3 \Rightarrow 5.8 \Rightarrow \boxed{x_1 = 0.533}$

$Q_2(8)$. $4x_1 + 6x_2 + 8x_3 = 0$

$6x_1 + 34x_2 + 52x_3 = -160$

$8x_1 + 52x_2 + 129x_3 = -452$

$$A = \begin{bmatrix} 4 & 6 & 8 \\ 6 & 34 & 52 \\ 8 & 52 & 129 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ -160 \\ -452 \end{bmatrix}$$

$A = L \cdot L^T$

$$A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} \Rightarrow \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{31}l_{21} + l_{22}l_{32} & l_{11}^2 + l_{22}^2 + l_{33}^2 \end{bmatrix}$$

- $l_{11}^2 = 4 \Rightarrow \boxed{l_{11} = 2}$
- $l_{11} l_{21} = 6 \Rightarrow \boxed{l_{21} = 3}$
- $l_{11} l_{31} = 8 \Rightarrow \boxed{l_{31} = 4}$
- $l_{21}^2 + l_{22}^2 \Rightarrow 34 \Rightarrow \boxed{l_{22} = 5}$
- $l_{21}l_{31} + l_{22}l_{32} = 52 \Rightarrow (3)(4) + (5)l_{32} = 52 \Rightarrow \boxed{l_{32} = 8}$
- $l_{11}^2 + l_{22}^2 + l_{33}^2 = 129 \Rightarrow (2)^2 + (5)^2 + l_{33}^2 = 129 \Rightarrow \boxed{l_{33} = 10}$

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Forward Substitutions $LY = b$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -160 \\ -452 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 8 & 10 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -160 \\ -452 \end{bmatrix}$$

$y_1 = 0$, $3(0) + 5y_2 = -160 \Rightarrow y_2 = -32$

$4(0) + 8(-32) + 10y_3 = -452 \Rightarrow y_3 = -19.6$

Backward Substitution: $L^T X = Y$

$$\begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -32 \\ -19.6 \end{bmatrix}$$

- $10x_3 = -19.6 \Rightarrow \boxed{x_3 = -1.96}$
- $5x_2 + 8x_3 = -32. \Rightarrow \boxed{-3.264 = x_2}$
- $2x_1 - 3.264(3) + 4(-1.96) = 0$ $\boxed{x_1 = 8.816}$

Q3(9): $0.01x_1 + \qquad + 0.03x_3 = 0.14$

$0.16x_2 + 0.08x_3 = 0.16$

$0.03x_1 + 0.08x_2 + 0.14x_3 = 0.54$

Sol

$$A = \begin{bmatrix} 0.01 & 0 & 0.03 \\ 0 & 0.16 & 0.08 \\ 0.03 & 0.08 & 0.14 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 0.14 \\ 0.16 \\ 0.54 \end{bmatrix}$$

$A = L \cdot L^T$

$$A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} \Rightarrow \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31} + l_{22}l_{32} & \cancel{l_{11}^2 + l_{22}^2 + l_{33}^2} \end{bmatrix}$$

$l_{31}^2 + l_{32}^2 + l_{33}^2$

Comparing.

- $l_{11}^2 = 0.01 \Rightarrow \boxed{l_{11} = 0.1}$ • $l_{11}l_{31} = 0.03 \Rightarrow \boxed{l_{31} = 0.03/0.1 = 0.3}$
- $l_{11}l_{21} = 0 \Rightarrow \boxed{l_{21} = 0}$ • $l_{21}^2 + l_{22}^2 = 0.16 \Rightarrow \boxed{l_{22} = 0.4}$
- $l_{21}l_{31} + l_{22}l_{32} = 0.08 \Rightarrow l_{32} = \frac{0.08}{0.4} \Rightarrow \boxed{l_{32} = 0.2}$
- $l_{31}^2 + l_{32}^2 + l_{33}^2 = 0.14 \Rightarrow \boxed{l_{33} = 0.1}$

Forward Substitution: $LY = b$.

$$\begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0.3 & 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.16 \\ 0.54 \end{bmatrix}$$

$\Rightarrow 0.1y_1 = 0.14 \Rightarrow \boxed{y_1 = 1.4}$

$\Rightarrow 0.4y_2 = 0.16 \Rightarrow \boxed{y_2 = 0.4}$

$\Rightarrow 0.3y_1 + 0.2y_2 + 0.1y_3 = 0.54$

$\boxed{y_3 = 0.4}$

Backward Substitution: $L^T X = Y$

$$\begin{bmatrix} 0.1 & 0 & 0.3 \\ 0 & 0.4 & 0.2 \\ 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.4 \\ 0.4 \\ 0.4 \end{bmatrix}$$

$\Rightarrow 0.1x_3 = 0.4 \Rightarrow \boxed{x_3 = 4}$

$\Rightarrow 0.4x_2 + 0.2x_3 = 0.4 \quad \boxed{x_2 = -1}$

$\Rightarrow 0.1x_1 + 0.3x_3 = 1.4 \quad \boxed{x_1 = 2}$

$Q_3(10)$. $4x_1 + \qquad + 2x_3 = 1.5$

$4x_2 + x_3 = 4.0$

$2x_1 + x_2 + 2x_3 = 2.5$

Sol

$$A = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 4 & 1 \\ 2 & 1 & 2 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 1.5 \\ 4.0 \\ 2.5 \end{bmatrix}$$

$A = L \cdot L^T$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31} + l_{22}l_{32} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

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- $l_{11}^2 = 4 \Rightarrow \boxed{l_{11}=2}$, $l_{11}l_{21} = 0 \Rightarrow \boxed{l_{21}=0}$, $l_{11}l_{31}=2 \Rightarrow \boxed{l_{31}=\frac{2}{2}=1}$
- $l_{21}^2+l_{22}^2 \Rightarrow 4 \Rightarrow 0+l_{22}^2=4 \Rightarrow \boxed{l_{22}=2}$
- $l_{21}l_{31}^{0}+l_{22}l_{32}=1 \rightarrow \boxed{l_{32}=0.5}$
- $l_{31}^2+l_{32}^2+l_{33}^2=2 \Rightarrow 1+(0.5)^2+l_{33}^2=2 \Rightarrow \boxed{0.866=l_{33}}$

Forward Substitution: $LY=b$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0.5 & 0.866 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}=\begin{bmatrix} 1.5 \\ 4.0 \\ 2.5 \end{bmatrix}$$

$\Rightarrow 2y_1=1.5$ $\boxed{y_1=0.75}$

$\Rightarrow \boxed{y_2=2}$

$\Rightarrow y_1+0.5y_2+0.866y_3=2.5$

$\boxed{y_3=0.866}$

Backward Substitution: $L^TX=Y$

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0.5 \\ 0 & 0 & 0.866 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0.75 \\ 2 \\ 0.866 \end{bmatrix}$$

$\boxed{x_3=1}$

$2x_2+0.5x_3=2$ $\boxed{x_2=0.75}$

$2x_1+x_3=0.75$ $\boxed{x_1=-0.125}$

$Q_4(ii)$: $x_1-x_2+3x_3+2x_4=15$ wrong

$-x_1+5x_2-5x_3-2x_4=-35$

$3x_1-5x_2+19x_3+3x_4=94$

$2x_1-2x_2+3x_3+21x_4=1$

$$A=\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix}\begin{bmatrix} l_{11} & l_{21} & l_{31} & l_{41} \\ 0 & l_{22} & l_{32} & l_{42} \\ 0 & 0 & l_{33} & l_{43} \\ 0 & 0 & 0 & l_{44} \end{bmatrix}$$

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• $\boxed{x_4 = -1}$

• $3x_3 - x_4 = 13 \Rightarrow 3x_3 + 1 \Rightarrow 13 \rightarrow 3x_3 = 12 \rightarrow \boxed{x_3 = 4}$

• $2x_2 - x_3 = 10 \rightarrow 2x_2 - 4 = 10 \rightarrow \boxed{x_2 = 3}$

• $x_1 - x_2 + 3x_3 + 4x_4 = 15$

$x_1 + 3 + 3(4) + 4(-1) = 15$

$\boxed{x_1 = }$ 2

Q_{12} (4).
$$\begin{aligned} 4x_1 + 2x_2 + 4x_3 \quad &= 20 \\ 2x_1 + 2x_2 + 3x_3 + 2x_4 &= 36 \\ 4x_1 + 3x_2 + 6x_3 + 3x_4 &= 60 \\ 2x_3 + 3x_3 + 9x_4 &= 122. \end{aligned}$$

$$\begin{bmatrix} l_{11} & l_{12} & l_{13} & l_{14} \\ & l_{22} & l_{23} & l_{24} \\ & & & \\ & & & \end{bmatrix}$$

Sol.

$A = L \cdot L^T$

$$A = \begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} & l_{41} \\ 0 & l_{22} & l_{32} & l_{42} \\ 0 & 0 & l_{33} & l_{43} \\ 0 & 0 & 0 & l_{44} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} & l_{11}l_{41} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} & l_{21}l_{41} + l_{22}l_{42} \\ l_{11}l_{31} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 & l_{31}l_{41} + l_{32}l_{42} + l_{33}l_{43} \\ l_{11}l_{41} & l_{41}l_{21} + l_{42}l_{22} & l_{41}l_{31} + l_{42}l_{32} + l_{43}l_{33} & l_{41}^2 + l_{42}^2 + l_{43}^2 + l_{44}^2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 2 & 4 & 0 \\ 2 & 2 & 3 & 3 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & 9 \end{bmatrix}$$

• $l_{11}^2 = 4 \Rightarrow \boxed{l_{11} = 2}$ • $l_{11}l_{41} = 0$

• $l_{11}l_{21} = 2 \Rightarrow \boxed{l_{21} = 1}$ $l_{41} = 0$

• $l_{11}l_{31} = 4 \rightarrow \boxed{l_{31} = 2}$

• $l_{21}^2 + l_{22}^2 = 2 \Rightarrow 1 + l_{22}^2 = 2 \rightarrow \boxed{l_{22} = 1}$

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- $l_{31}l_{21} + l_{32}l_{22} = 3 \Rightarrow 2 + l_{32} = 3 \Rightarrow \boxed{l_{32} = 1}$
- $l_{41}l_{21}^{\,0} + l_{42}l_{22} = 2 \Rightarrow \boxed{l_{42} = 2}$
- $l_{31}^2 + l_{32}^2 + l_{33}^2 = 6 \Rightarrow 4 + 1 + l_{33}^2 = 6 \Rightarrow \boxed{l_{33} = 1}$
- $l_{41}l_{31}^{\,0} + l_{42}l_{32} + l_{43}l_{33} = 3 \Rightarrow 2 + l_{43} = 3 \Rightarrow \boxed{l_{43} = 1}$
- $l_{41}^2 + l_{42}^2 + l_{43}^2 + l_{44}^2 = 9 \Rightarrow 0 + 4 + 1 + l_{44}^2 = 9 \Rightarrow \boxed{l_{44} = 2}$

Forward Substitution: $L \cdot Y = b$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 0 & 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 20 \\ 36 \\ 60 \\ 122 \end{bmatrix}$$

- $2y_1 = 20 \Rightarrow \boxed{y_1 = 10}$
- $y_1 + y_2 = 36 \Rightarrow y_2 = 36 - 10$
  $\boxed{y_2 = 26}$
- $2y_1 + y_2 + y_3 = 60 \Rightarrow 2(10) + (26) + y_3 = 60 \Rightarrow \boxed{y_3 = 14}$
- $2y_2 + y_3 + 2y_4 = 122 \Rightarrow 2(26) + (14) + 2y_4 = 122 \Rightarrow \boxed{28 = y_4}$

Backward Substitution: $L^T X = Y$

$$\begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 26 \\ 14 \\ 28 \end{bmatrix}$$

- $2x_4 = 28 \Rightarrow \boxed{x_4 = 14}$
- $x_3 + x_4 = 14 \Rightarrow \boxed{x_3 = 0}$
- $x_2 + x_3 + 2x_4 = 26$
  $\boxed{x_2 = -2}$
- $2x_1 + x_2 + 2x_3 = 10 \Rightarrow \boxed{x_1 = 6}$

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CHEBYSHEV METHOD:

Q1. Find the roots of $f(x) = x^3 - 2x - 5$ with initial given $x_0 = 2$ correct upto 3 decimal places.

$f(x) = x^3 - 2x - 5$ $f'(x) = 3x^2 - 2$ $f''(x) = 6x$

$f(x_0) = -1$ $f'(x_0) = 10$ $f''(x_0) = 12$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} - \frac{f''(x_i)}{2f'(x_i)}\left(\frac{f(x_i)}{f'(x_i)}\right)^2$$

Ob	x_i	$f(x_i)$	$f'(x_i)$	$f''(x_i)$		
0	2	-1	10	12	2.094	
1	2.094	-0.0062	11.1545	12.564	2.094	

Q2. $f(x) = e^{-x} - x$. $x_0 = 0.5$, Perform 3 iterations.

$f(x) = e^{-x} - x$ $f'(x) = -e^{-x} - 1 \Rightarrow f''(x) = +e^{-x}$

$f(0.5) = 0.10653$ $f'(0.5) = -1.60653$ $f''(0.5) = 0.60653$

0	0.5	0.10653	-1.60653	0.60653	0.56714
1	0.56714	5.15654×10^{-6}	-1.56714	0.56714	0.56714
2	0.56714	" "	" "	" "	0.56714